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%Ravi Pratap Singh
%M2024BSASS022
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```
AllahabadFinalTotalUrbanRuralwithDistrict
```

```
AllahabadFinalTotalUrbanRuralwithDistrict = 3x87 table
```

	District	TRU	No_HH	TOT_P	TOT_M	TOT_F	P_06	M_06
1	Allahabad	"Rural"	721011	4481518	2340959	2140559	722406	380936
2	Allahabad	"Total"	750586	4646659	2428707	2217952	743028	391900
3	Allahabad	"Urban"	29575	165141	87748	77393	20622	10964

```
% Load Excel file
T = AllahabadFinalTotalUrbanRuralwithDistrict;
% Preview data
head(T)
```

```
ans = 3x87 table
```

	District	TRU	No_HH	TOT_P	TOT_M	TOT_F	P_06	M_06
1	Allahabad	"Rural"	721011	4481518	2340959	2140559	722406	380936
2	Allahabad	"Total"	750586	4646659	2428707	2217952	743028	391900
3	Allahabad	"Urban"	29575	165141	87748	77393	20622	10964

```
% Population
TOT_P = T.TOT_P;
TOT_M = T.TOT_M;
TOT_F = T.TOT_F;

% Education
P_LIT = T.P_LIT;
M_LIT = T.M_LIT;
F_LIT = T.F_LIT;

% Employment
TOT_WORK = T.TOT_WORK_P;
MAIN_WORK = T.MAINWORK_P;
MARG_WORK = T.MARGWORK_P;
NON_WORK = T.NON_WORK_P;

% Social groups
SC_P = T.P_SC;
ST_P = T.P_ST;

% Children (Health/Demographic proxy)
P_06 = T.P_06;
```

```
Literacy_Rate = (P_LIT ./ TOT_P) * 100;
```

```
Sex_Ratio = (TOT_F ./ TOT_M) * 1000;
```

```
WPR = (TOT_WORK ./ TOT_P) * 100;
```

```
Main_Work_Share = (MAIN_WORK ./ TOT_WORK) * 100;  
Marg_Work_Share = (MARG_WORK ./ TOT_WORK) * 100;
```

```
Child_Dependency = (P_06 ./ TOT_P) * 100;
```

```
SC_Share = (SC_P ./ TOT_P) * 100;  
ST_Share = (ST_P ./ TOT_P) * 100;
```

```
mean_LR = mean(Literacy_Rate);  
std_LR = std(Literacy_Rate);
```

```
mean_WPR = mean(WPR);  
std_WPR = std(WPR);
```

```
% Remove missing values if any
```

```
X = Literacy_Rate;  
Y = WPR;
```

```
% Pearson correlation (manual)
```

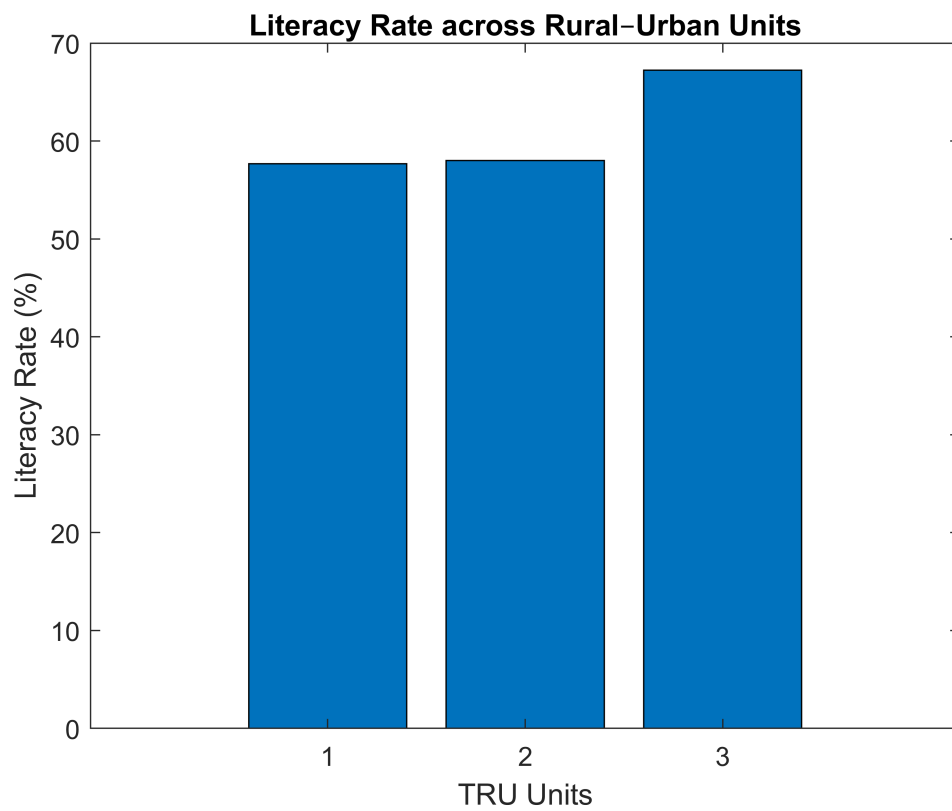
```
X_mean = mean(X);  
Y_mean = mean(Y);
```

```
numerator = sum((X - X_mean) .* (Y - Y_mean));  
denominator = sqrt(sum((X - X_mean).^2) * sum((Y - Y_mean).^2));
```

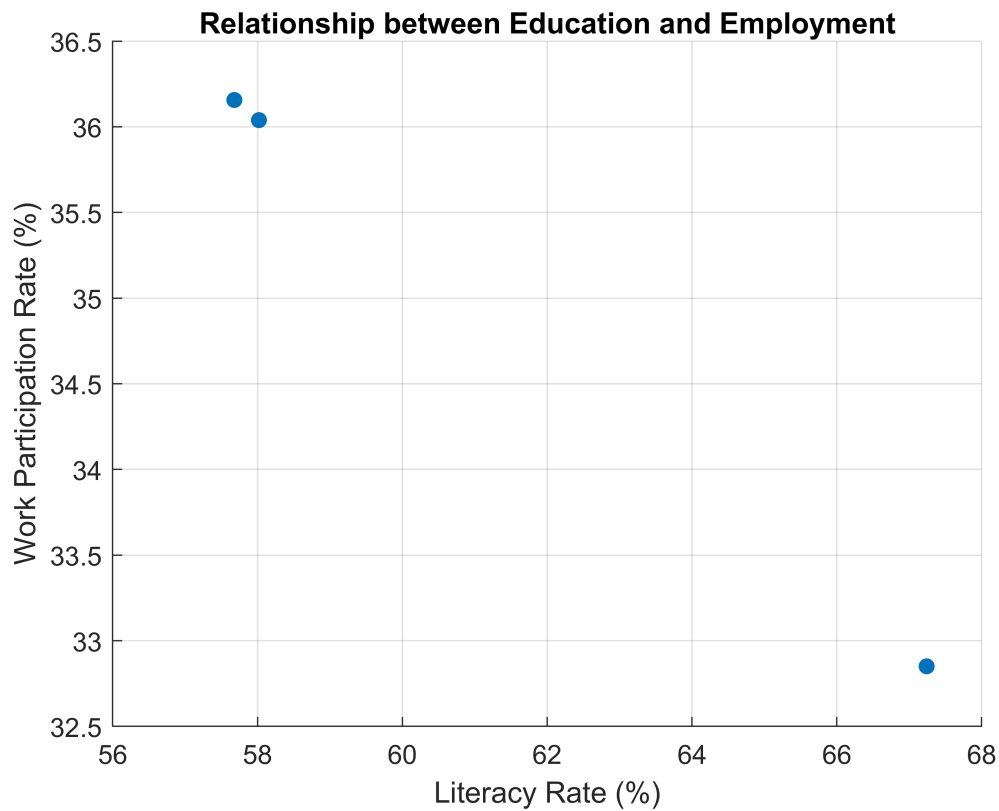
```
corr_Lit_WPR = numerator / denominator;
```

```
Gender_Lit_Gap = M_LIT - F_LIT;  
Gender_Work_Gap = T.TOT_WORK_M - T.TOT_WORK_F;
```

```
figure  
bar(Literacy_Rate)  
xlabel('TRU Units')  
ylabel('Literacy Rate (%)')  
title('Literacy Rate across Rural-Urban Units')
```



```
figure
scatter(Literacy_Rate, WPR, 'filled')
xlabel('Literacy Rate (%)')
ylabel('Work Participation Rate (%)')
title('Relationship between Education and Employment')
grid on
```



```
Z_Literacy = (Literacy_Rate - mean(Literacy_Rate)) / std(Literacy_Rate);
Z_WPR      = (WPR - mean(WPR)) / std(WPR);
```

```
% Regression using normal equation
X = [ones(length(Literacy_Rate),1), Literacy_Rate];
Y = WPR;

beta = (X' * X) \ (X' * Y);

Intercept = beta(1);
Slope     = beta(2);
```

```
CV_Literacy = (std(Literacy_Rate) / mean(Literacy_Rate)) * 100;
CV_WPR      = (std(WPR) / mean(WPR)) * 100;
```

```
Development_Index = (Z_Literacy + Z_WPR) / 2;
```

```
% Correlation-based t-test (manual, no toolbox)

r = corr_Lit_WPR;           % Pearson correlation
n = length(Literacy_Rate);
```

```

% t-statistic formula
t_stat = r * sqrt((n - 2) / (1 - r^2));

% degrees of freedom
df = n - 2;

% critical value for alpha = 0.05 (two-tailed), df = 1
t_critical = 4.303;

% hypothesis decision
if abs(t_stat) > t_critical
    decision = "Reject H0: Relationship is statistically significant";
else
    decision = "Fail to reject H0: Relationship is not statistically significant";
end

% display results
t_stat

```

```
t_stat = -Inf
```

```
df
```

```
df = 1
```

```
decision
```

```

decision =
"Reject H0: Relationship is statistically significant"

```

```

Rural_Lit = Literacy_Rate(T.TRU == "Rural");
Urban_Lit = Literacy_Rate(T.TRU == "Urban");
mean_R = mean(Rural_Lit);
mean_U = mean(Urban_Lit);

var_R = var(Rural_Lit);
var_U = var(Urban_Lit);

nR = length(Rural_Lit);
nU = length(Urban_Lit);

t_ru = (mean_R - mean_U) / sqrt((var_R/nR) + (var_U/nU));

```

```

figure
scatter(Literacy_Rate, WPR, 120, 'k', 'filled')
hold on

% Regression line
xfit = linspace(min(Literacy_Rate), max(Literacy_Rate), 100);
yfit = Intercept + Slope * xfit;
plot(xfit, yfit, 'k-', 'LineWidth', 2)

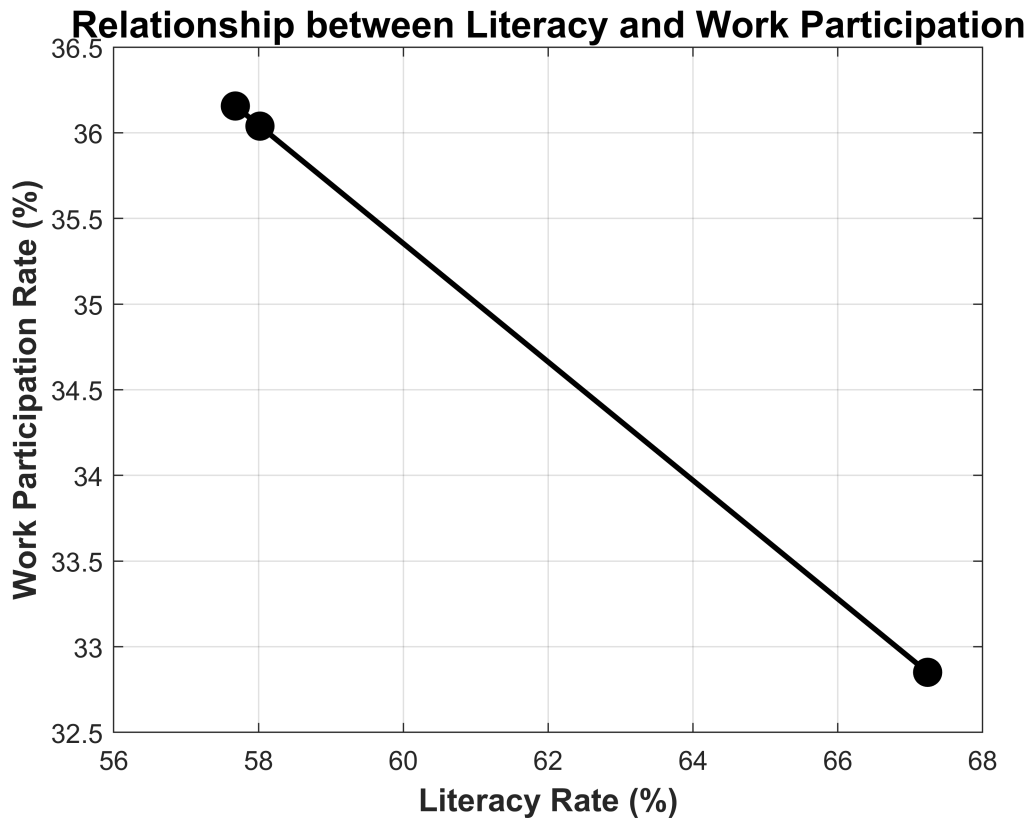
xlabel('Literacy Rate (%)', 'FontSize', 12, 'FontWeight', 'bold')

```

```

ylabel('Work Participation Rate (%)', 'FontSize', 12, 'FontWeight', 'bold')
title('Relationship between Literacy and Work Participation', 'FontSize', 14)
grid on
box on

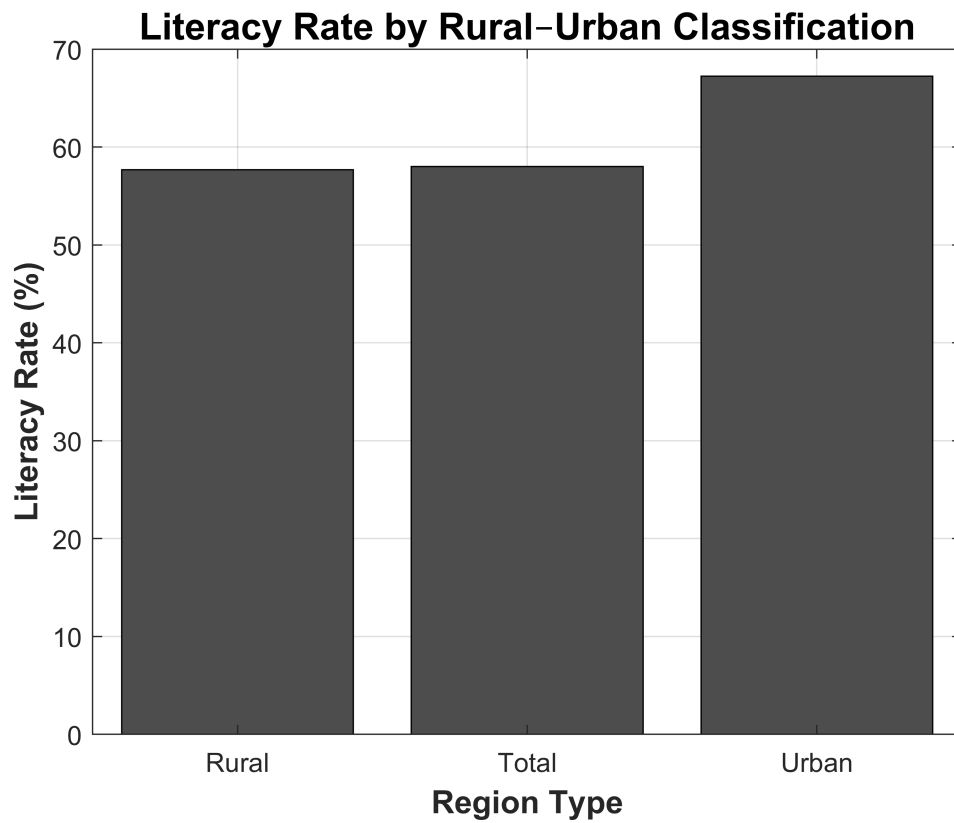
```



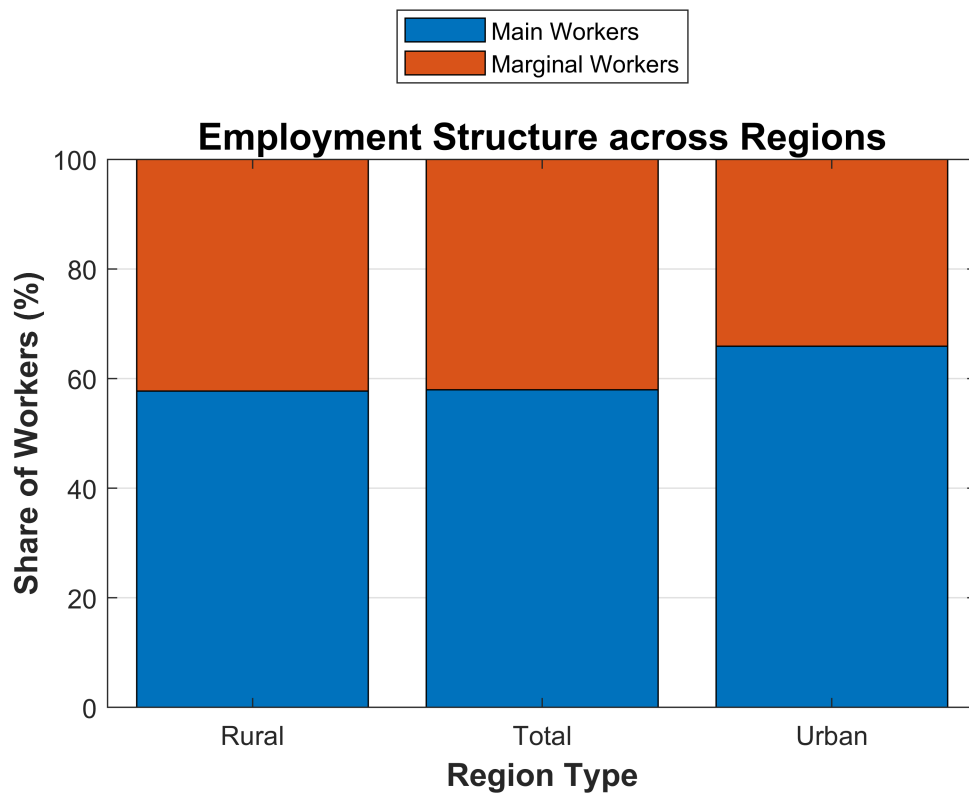
```

figure
bar(categorical(T.TRU), Literacy_Rate, 'FaceColor', [0.3 0.3 0.3])
xlabel('Region Type', 'FontSize', 12, 'FontWeight', 'bold')
ylabel('Literacy Rate (%)', 'FontSize', 12, 'FontWeight', 'bold')
title('Literacy Rate by Rural-Urban Classification', 'FontSize', 14)
grid on
box on

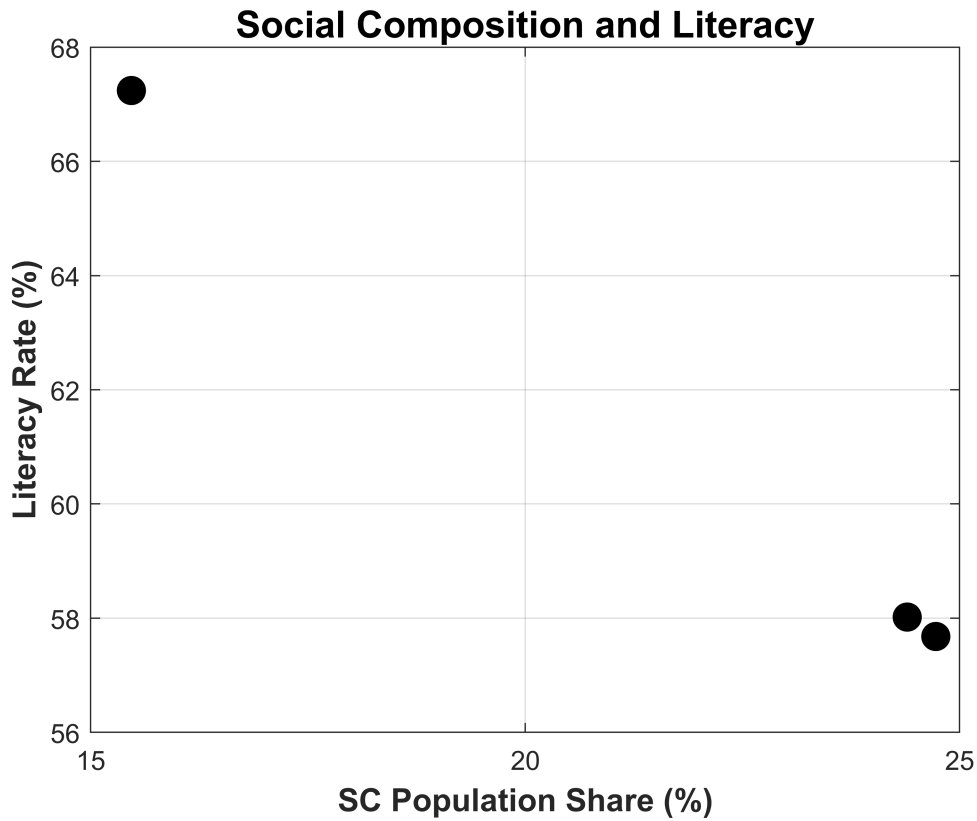
```



```
figure
bar(categorical(T.TRU), [Main_Work_Share Marg_Work_Share], 'stacked')
xlabel('Region Type','FontSize',12,'FontWeight','bold')
ylabel('Share of Workers (%)','FontSize',12,'FontWeight','bold')
legend({'Main Workers','Marginal Workers'}, 'Location','northoutside')
title('Employment Structure across Regions','FontSize',14)
grid on
box on
```



```
figure
scatter(SC_Share, Literacy_Rate, 120, 'k', 'filled')
xlabel('SC Population Share (%)', 'FontSize', 12, 'FontWeight', 'bold')
ylabel('Literacy Rate (%)', 'FontSize', 12, 'FontWeight', 'bold')
title('Social Composition and Literacy', 'FontSize', 14)
grid on
box on
```

```
T.Literacy_Rate = Literacy_Rate;
T.WPR = WPR;
T.Sex_Ratio = Sex_Ratio;
T.SC_Share = SC_Share;
T.ST_Share = ST_Share;
```

```
clc;
close all;
```

```
disp('TRU-WISE INDICATORS ')
```

TRU-WISE INDICATORS

```
disp(T(:, {'TRU', 'TOT_P', 'Literacy_Rate', 'WPR', 'Sex_Ratio', 'SC_Share', 'ST_Share'}))
```

TRU	TOT_P	Literacy_Rate	WPR	Sex_Ratio	SC_Share	ST_Share
"Rural"	4.4815e+06	57.68	36.156	914.39	24.725	0.10847
"Total"	4.6467e+06	58.019	36.039	913.22	24.396	0.12801
"Urban"	1.6514e+05	67.24	32.851	881.99	15.468	0.65823

```
disp(' ')
```

```
disp('DESCRIPTIVE STATISTICS')
```

DESCRIPTIVE STATISTICS

```
disp(table(mean_LR, std_LR, CV_Literacy, mean_WPR, std_WPR, CV_WPR, ...  
          'VariableNames',{ 'Mean_LR','SD_LR','CV_Literacy','Mean_WPR','SD_WPR','CV_WPR'})))
```

Mean_LR	SD_LR	CV_Literacy	Mean_WPR	SD_WPR	CV_WPR
60.98	5.4244	8.8954	35.015	1.8754	5.3561

```
disp(' ')
```

```
disp('CORRELATION & HYPOTHESIS TEST ')
```

CORRELATION & HYPOTHESIS TEST

```
disp(table(r, t_stat, df, t_critical, decision, ...  
          'VariableNames',{ 'Correlation_r','t_statistic','df','t_critical','Decision'})))
```

Correlation_r	t_statistic	df	t_critical	Decision
-1	-Inf	1	4.303	"Reject H0: Relationship is statistically significant"

```
disp(' ')
```

```
disp('REGRESSION RESULTS')
```

REGRESSION RESULTS

```
disp(table(Intercept, Slope, ...  
          'VariableNames',{ 'Intercept','Slope'})))
```

Intercept	Slope
56.098	-0.34574

```
disp(' ')
```

```
disp('KEY SOCIO-ECONOMIC INDICATORS ')
```

KEY SOCIO-ECONOMIC INDICATORS

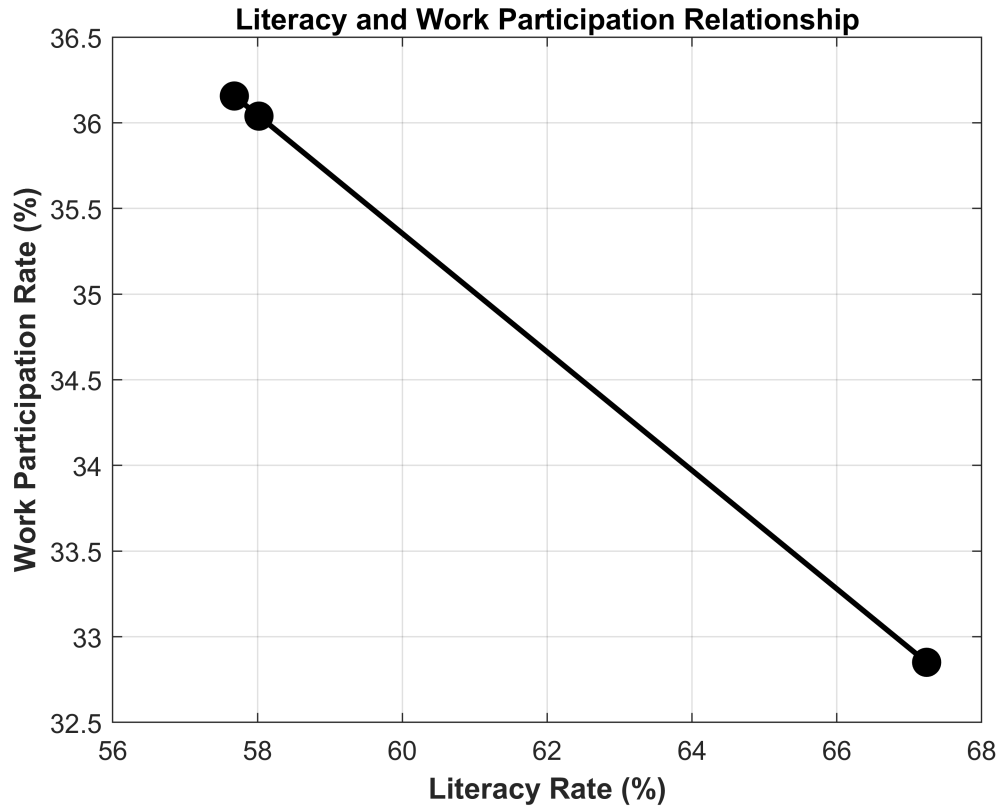
```
disp(table(Literacy_Rate, WPR, Sex_Ratio, SC_Share, ST_Share, Child_Dependency))
```

Literacy_Rate	WPR	Sex_Ratio	SC_Share	ST_Share	Child_Dependency
---------------	-----	-----------	----------	----------	------------------

57.68	36.156	914.39	24.725	0.10847	16.12
58.019	36.039	913.22	24.396	0.12801	15.991
67.24	32.851	881.99	15.468	0.65823	12.488

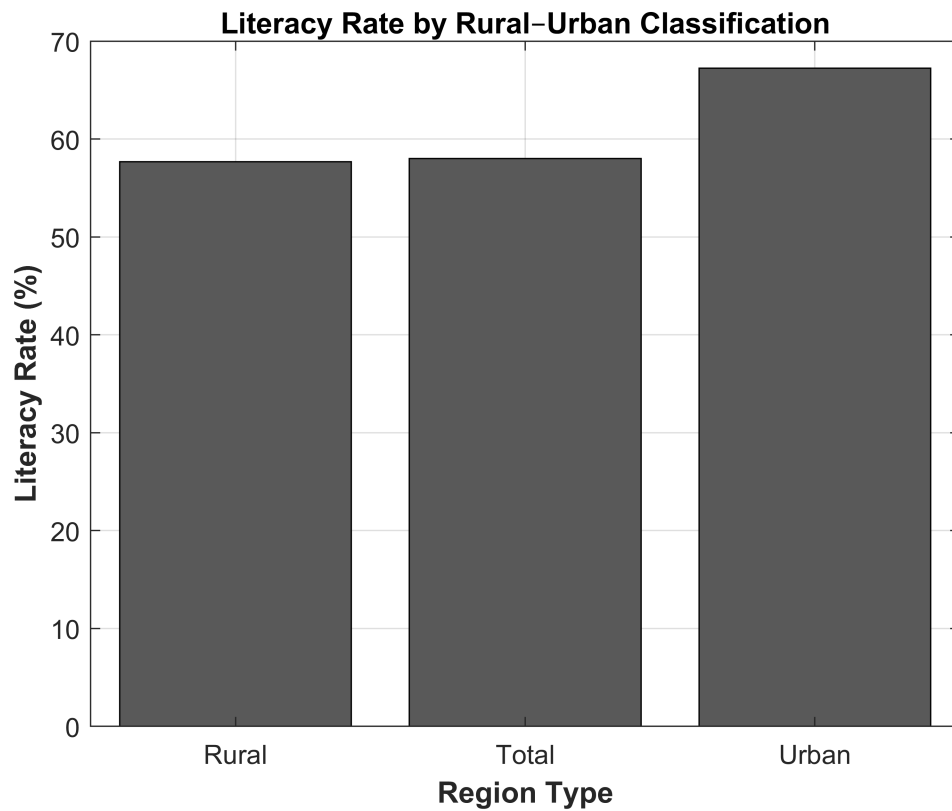
% 1. Literacy vs Work Participation (Core Study Graph)

```
figure
scatter(Literacy_Rate, WPR, 120, 'k', 'filled')
hold on
xfit = linspace(min(Literacy_Rate), max(Literacy_Rate), 100);
yfit = Intercept + Slope * xfit;
plot(xfit, yfit, 'k-', 'LineWidth', 2)
xlabel('Literacy Rate (%)', 'FontWeight', 'bold')
ylabel('Work Participation Rate (%)', 'FontWeight', 'bold')
title('Literacy and Work Participation Relationship')
grid on; box on
```



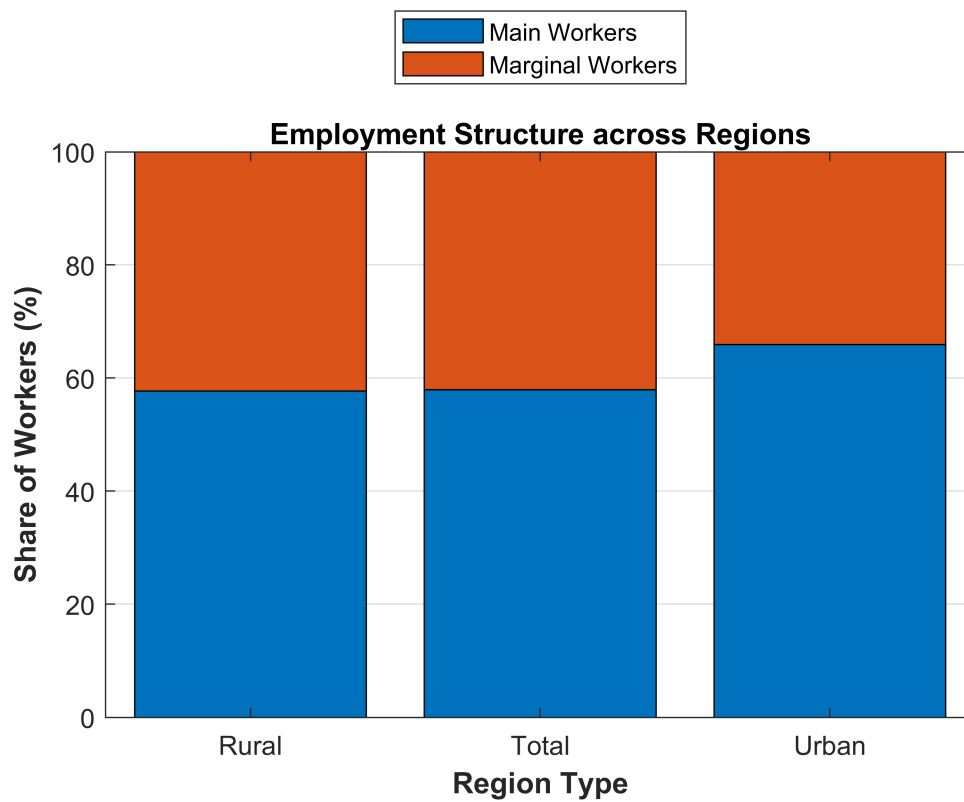
% 2. Rural-Urban-Total Literacy Comparison

```
figure
bar(categorical(T.TRU), Literacy_Rate, 'FaceColor', [0.35 0.35 0.35])
xlabel('Region Type', 'FontWeight', 'bold')
ylabel('Literacy Rate (%)', 'FontWeight', 'bold')
title('Literacy Rate by Rural-Urban Classification')
grid on; box on
```



% 3. Employment Structure: Main vs Marginal Workers

```
figure
bar(categorical(T.TRU), [Main_Work_Share Marg_Work_Share], 'stacked')
xlabel('Region Type','FontWeight','bold')
ylabel('Share of Workers (%)','FontWeight','bold')
legend({'Main Workers','Marginal Workers'},'Location','northoutside')
title('Employment Structure across Regions')
grid on; box on
```



% 4. Social Composition vs Literacy

```
figure
scatter(SC_Share, Literacy_Rate, 120, 'k', 'filled')
xlabel('SC Population Share (%)', 'FontWeight', 'bold')
ylabel('Literacy Rate (%)', 'FontWeight', 'bold')
title('Caste Composition and Literacy')
grid on; box on
```

